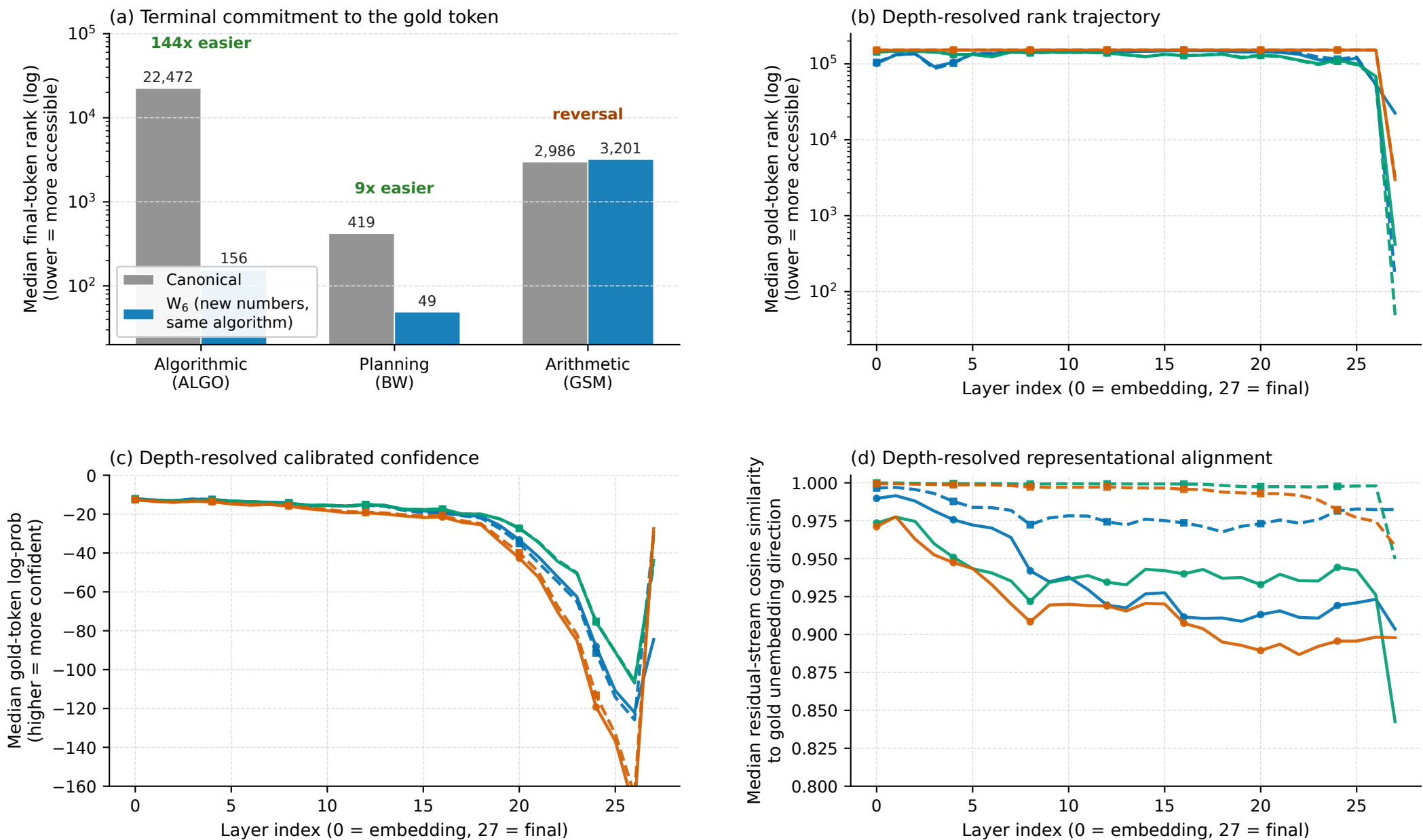


Mechanistic dissociation (Qwen-2.5-7B residual stream): W_6 surfaces the gold token across all four depth-resolved metrics for algorithmic and planning families, but not for arithmetic



Qwen-2.5-7B, 28 layers. Algorithmic: $n_{can}=100/n_{W6}=90$, Planning: $n_{can}=65/n_{W6}=65$, Arithmetic: $n_{can}=44/n_{W6}=24$. Coin Change omitted (no W_6 sample).

Algorithmic (ALGO) Planning (BW) Arithmetic (GSM) Canonical W_6